

PROJECT DESIGN REQUIREMENT

PDR — Version 1.0

Cute E-Commerce Platform

Keychains | Bouquets | Bags

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Prepared By	Development Team
Client	Cute E-Commerce Store Owner

1. Executive Summary

This document defines the full Product Design Requirements (PDR) for a cute, high-performance e-commerce platform selling Keychains, Bouquets, and Bags. The platform is designed to deliver a Myntra/Ajio-grade shopping experience layered with playful, kawaii-inspired visual design, buttery smooth interactions, and zero-lag performance.

The client's primary mandate is that every page should feel alive, cute, and interactive. Products must be front and centre at all times, and the user must never feel any friction — especially during add-to-cart and checkout flows.

2. Project Overview

2.1 Project Identity

Project Name	Cute E-Commerce Platform
Product Type	Full-Stack E-Commerce Web Application
Business Model	B2C Direct-to-Consumer
Target Market	India (Primary) — Young Adults, Gift Buyers, Students
Product Categories	Keychains, Bouquets, Bags
Platform	Web (Desktop + Mobile Responsive)
Launch Target	TBD by client

2.2 Design Philosophy

The design language is anchored in Pastel Kawaii — soft pinks, lilac purples, mint greens, and warm creams. Think Sanrio meets a boutique Etsy shop. Every interaction should delight the user — rounded corners, bouncy micro-interactions, floating particles, and animated feedback on every meaningful action.

Layout inspiration is drawn from Myntra and Ajio — product-first, clean navigation, editorial banners, and a sticky category filter that feels effortless.

3. Technology Stack

Layer	Technology	Purpose
Frontend Framework	Next.js 14 (App Router)	SSR, routing, performance, SEO
Styling	Tailwind CSS + shadcn/ui	Utility-first styling, pre-built UI components
3D & Animations	Three.js + Framer Motion + GSAP	3D scenes, spring animations, scroll triggers
Lottie	Lottie React	Delivery boy checkout animation
State Management	Zustand	Cart, wishlist, UI state — instant updates
Backend / Auth	Supabase	Auth, Realtime DB, Storage, Row Level Security
Database	Supabase PostgreSQL + Prisma ORM	Relational data, type-safe queries
Email Service	Resend	Transactional emails — OTP, order confirmation
Payments	Razorpay	INR payments, UPI, cards, wallets
Forms & Validation	React Hook Form + Zod	Client-side form handling and schema validation
Data Fetching	React Query (TanStack)	Caching, background sync, skeleton states
Notifications	Sonner	Toast notifications for all user actions
Icons	Lucide React + React Icons	Consistent icon system
SEO	Next.js Metadata API	Per-page SEO, Open Graph, structured data
Deployment	Vercel + Supabase Cloud	CI/CD, edge functions, global CDN

4. Site Architecture & Pages

4.1 Page Map

Route	Page Name	Access Level
/	Home Page	Public
/products	All Products (filtered)	Public
/products/[slug]	Product Detail Page	Public
/cart	Shopping Cart	Public (requires auth to checkout)
/checkout	Checkout — Multi-step	Authenticated
/checkout/success	Order Success + Delivery Animation	Authenticated
/account	User Dashboard	Authenticated
/account/orders	Order History & Tracking	Authenticated
/admin	Admin Dashboard	Admin Role Only
/admin/products	Product Management	Admin Role Only
/admin/orders	Order Management	Admin Role Only
/admin/users	User Management	Admin Role Only
/admin/coupons	Coupon & Discount Manager	Admin Role Only
/admin/settings	Store Settings	Admin Role Only
/admin/analytics	Analytics Dashboard	Admin Role Only

4.2 Home Page — Layout Specification

The home page follows a Myntra/Ajio-style product-first layout:

- Top Navigation Bar: Logo | Live Search Bar | Wishlist icon | Cart icon with badge | Account icon
- Announcement Marquee: Scrolling text — free delivery threshold, new arrivals, discount codes
- Hero Banner: Full-width editorial banner with 3 rotating slides (autoplay + dot nav). No Three.js here — keep it fast and clean.
- Category Pill Row: [All] [Bouquets] [Keychains] [Bags] — horizontal pill tabs, sticky below nav on scroll
- Product Grid: Ajio-style 4-column grid (desktop) / 2-column (mobile), tall portrait product cards
- Filter & Sort Bar: Above product grid — Sort dropdown + Filter drawer (price range, tags, availability)

- Why Us Row: Icons — Handmade with Love, Fast Delivery, Gift Wrapping, Custom Orders
- Testimonials: Soft card carousel with verified buyer badges
- Footer: Links, social icons, newsletter OTP signup, payment method logos

4.3 Category Filter Logic

Category Tab Behaviour

Clicking a category tab (All / Bouquets / Keychains / Bags) instantly filters the product grid WITHOUT a full page reload. State is managed via Zustand + React Query. Each category is pre-fetched on tab hover so switching feels instantaneous. The URL updates to `/?category=bouquets` for shareability and SEO. Active tab has a filled pastel pink background; inactive tabs are outlined.

4.4 Product Card Specification (Myntra-style)

Every product card must include the following elements:

- Tall portrait image (3:4 aspect ratio) — primary product photo
- Wishlist heart icon — top-right corner, toggles with pop animation
- Discount badge — bottom-left of image (e.g. 42% OFF)
- Bestseller / New badge — top-left ribbon if applicable
- Product name — below image, 1-2 lines max with ellipsis
- Price row — Sale price bold, original price strikethrough, discount % in green
- Star rating + review count
- Add to Cart — appears on hover (desktop) / always visible (mobile), slides up from card bottom with confetti burst on click

5. Checkout & Delivery Animation

5.1 Checkout Flow — Multi-Step

Step	Screen	Details
Step 1	Address	Saved addresses or new address form. Autofill via browser.
Step 2	Payment	Razorpay embedded — UPI, cards, net banking, wallets
Step 3	Review	Order summary before final confirmation
Post-Payment	Delivery Animation Page	Full-screen Lottie scene (see below)

5.2 Delivery Animation — Checkout Success (KEY FEATURE)

Animation Sequence

1. Cute delivery boy character walks in from the right carrying a package. 2. He opens the back of a pastel delivery truck. 3. Gently places the package inside. 4. Closes the door. 5. Waves to the camera. 6. Truck drives off to the left with a cheerful honk. 7. Screen fades to order confirmation with: Order ID, items ordered, estimated delivery date, and a 'Track your order' button.

Implementation: Lottie React with a custom JSON animation file. The animation plays once and auto-transitions to the confirmation screen. An order confirmation email is simultaneously dispatched via Resend.

6. Authentication System

6.1 Login Methods

- Email OTP (Magic Link) — Supabase Auth. User enters email, receives a 6-digit OTP, enters it, session created.
- Phone OTP — Supabase SMS Auth. User enters phone number, receives SMS OTP, enters 6-digit code.
- Google OAuth — One-click Google sign-in via Supabase OAuth provider.
- Guest browsing is allowed. Cart is saved to localStorage for guests.
- On login, guest cart is merged with the user's saved cart in Supabase.

6.2 Admin Authentication

- Separate login route: /admin/login
- Role check via Supabase Row Level Security (RLS): role = 'admin' in users table
- All admin routes are server-protected — unauthorized access redirects to home

7. Email & Notification System

Email Type	Trigger	Content
OTP Verification	User requests login OTP	Brand header, 6-digit code, 10-minute expiry
Welcome Email	First signup/login	Brand intro, featured products, discount code
Order Confirmation	Payment successful	Order items, total, delivery estimate, Track button
Status Update	Admin changes order status	New status, order details, support contact
Newsletter Confirm	User subscribes from footer	Thank you + OTP confirmation of email

8. Admin Panel — Full Control

The Admin Panel gives the store owner god-mode access to every single aspect of the platform. All admin routes are protected behind Supabase RLS role checks.

8.1 Admin Modules

Module	Capabilities
Dashboard	Revenue chart (Recharts), total orders, active users, GMV, low stock alerts, today's orders
Products	Add / Edit / Delete products, upload images to Supabase Storage, manage variants, set featured flag
Categories	Manage Keychains / Bouquets / Bags — add new categories, edit icons and banners
Orders	View all orders, update status (Processing → Packed → Shipped → Delivered), trigger email on change
Users	View all registered users, see order history per user, ban/unban accounts
Coupons	Create discount codes, set discount type (% or flat), expiry date, usage limit, track usage
Analytics	Page views, conversion funnel, top products, revenue by category, repeat purchase rate
Email Templates	Preview and live-edit transactional email HTML templates
Settings	Store name, logo, announcement bar text, social links, homepage banner images, footer content

9. Database Schema

9.1 Core Tables

Table	Key Fields
users	id, name, email, phone, role (user/admin), avatar_url, created_at
products	id, name, slug, description, price, mrp, category_id, stock, images[], tags[], is_featured, is_active
categories	id, name, slug, icon_url, banner_image_url, sort_order
orders	id, user_id, items (JSONB), subtotal, discount, total, status, address_id, payment_id, created_at
order_items	id, order_id, product_id, quantity, price_at_purchase, variant

addresses	id, user_id, label, full_name, line1, line2, city, state, pincode, phone
cart	id, user_id, items (JSONB), updated_at
wishlist	id, user_id, product_ids[]
coupons	id, code, discount_type, value, min_order, expiry, usage_limit, used_count, is_active
reviews	id, product_id, user_id, rating (1-5), comment, is_verified, created_at

10. Performance & UX Requirements

10.1 No-Lag Standards

- Cart updates use Zustand optimistic UI — cart reflects instantly before Supabase sync confirms
- Category tab switching: products pre-fetched on hover via React Query — zero loading delay on click
- All product images served via Next.js Image Optimization with blur placeholder on load
- Skeleton loaders on every data-fetching route — no blank screens ever
- React Suspense boundaries throughout the app
- Next.js prefetch on hover for all product links
- Supabase Realtime for cart sync across browser tabs and admin order updates
- Error states: every async action has a retry-able error UI

10.2 Cute Interactions Checklist

- Custom sparkle/heart cursor following mouse movement
- Floating heart/star particles on product card hover
- Confetti burst animation on Add to Cart
- Bouncy spring animation on all CTA buttons (Framer Motion spring)
- 3D tilt card effect on product cards using Framer Motion useMotionValue
- Scroll-triggered stagger animations on every homepage section (GSAP ScrollTrigger)
- Animated number counters on admin dashboard stats
- Ribbon/badge entrance animations on Bestseller and New product cards
- Page transition animations using Framer Motion AnimatePresence
- Cute empty states — empty cart shows a sad plushie SVG, no search results shows a searching cat
- Wishlist heart — instant fill animation with a pop scale effect on toggle
- Three.js scenes scoped to: checkout success delivery animation + optional subtle footer accent

11. Folder Structure

<code>/app/(store)/page.tsx</code>	← Home Page
<code>/app/(store)/products/page.tsx</code>	← Products Grid
<code>/app/(store)/products/[slug]/page.tsx</code>	← Product Detail
<code>/app/(store)/cart/page.tsx</code>	← Cart
<code>/app/(store)/checkout/page.tsx</code>	← Checkout
<code>/app/(store)/checkout/success/page.tsx</code>	← Delivery Animation
<code>/app/(store)/account/page.tsx</code>	← User Dashboard

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/app/(admin)/admin/page.tsx      ← Admin Dashboard
/app/(admin)/admin/products/page.tsx ← Product CRUD
/app/(admin)/admin/orders/page.tsx ← Order Management
/components/ui/                  ← shadcn components
/components/three/               ← Three.js scenes
/components/animations/          ← Lottie + Framer
/lib/supabase.ts                 ← Supabase client
/lib/prisma.ts                   ← Prisma ORM client
/lib/resend.ts                   ← Resend email client
/lib/zustand/store.ts            ← Global state
/emails/                         ← React Email templates
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12. Deployment & Infrastructure

Frontend + Backend	Vercel (Next.js native deployment, edge functions, CDN)
Database	Supabase Hosted PostgreSQL
File Storage	Supabase Storage Buckets (product images, banners)
Email Delivery	Resend with custom domain DNS records
Payment Gateway	Razorpay (INR, UPI, cards, net banking)
Environment Vars	.env.local — Supabase URL/keys, Resend API key, Razorpay keys, OTP config

13. Acceptance Criteria

The project is considered ready for client handoff when ALL of the following are met:

- All 15 pages/routes are implemented and fully functional
- Category pill filter works instantly with zero lag on tab switch
- Add to Cart and Checkout flow work end-to-end without errors
- Email OTP and Phone OTP authentication work correctly
- Order confirmation email is received within 30 seconds of payment
- Delivery animation plays correctly on checkout success
- Admin can perform CRUD on products, categories, orders, coupons, and users
- All cute interactions (confetti, bouncy buttons, hover particles) are implemented
- Site is fully mobile responsive on iOS Safari and Android Chrome
- Page load time under 2 seconds on a standard 4G connection
- No broken links, missing images, or empty states without fallback UI

End of PDR Document — Version 1.0